● LIVE · portfolio walkthrough

Running a local-first support agent on a Mac.

How AssistSupport drafts KB-grounded IT support responses in under 25 ms — without a single query leaving the laptop.

Saagar Patel · IT Platform Eng · Local-first demo

IT support drowns in the same questions — and cloud AI isn't a clean fix.

- Every IT team repeats itself: ~25% of tickets are policy / howto questions already answered in the KB.
- Cloud LLMs promise automation but add three sharp costs: data leaves the tenant, hallucinations look confident, and per-seat pricing compounds.
- Vendor assistants hide their routing and retrieval — when they answer wrong, you can't debug why.
- The real bar: draft something a human would actually paste into Jira, cite where the claim came from, and be honest when the KB doesn't know.

A second brain — not a replacement.

01

LOCAL-FIRST

App, sidecar, classifier, retrieval, reranker, and LLM all run on-device in the demo path. SQLCipher AES-256 protects core workspace data at rest.

02

KB-GROUNDED

Every draft cites KB articles from the local index. Hybrid retrieval narrows candidates before rerank; inline [n] markers you can click.

03

TRUST-GATED

Confidence modes (answer / clarify / abstain). The model is allowed to refuse when the KB doesn't cover the question.

You don't need a foundation model on every desk. You need a pipeline that knows the KB cold, runs fast, and keeps the operator in the loop.

The pipeline — five stages, all local.

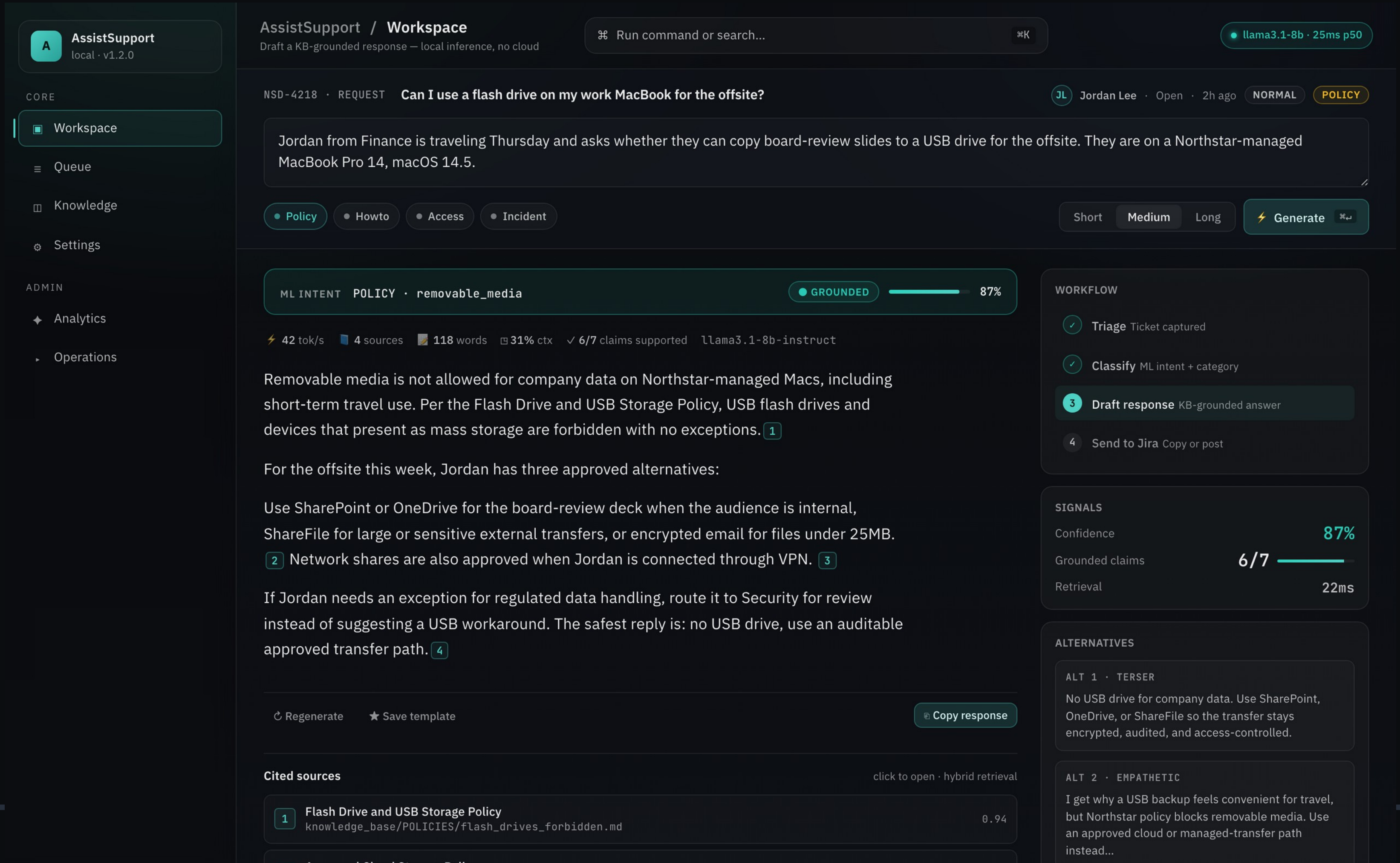
INTENT	RETRIEVE	RERANK	DRAFT	LEARN
logreg	TF-IDF	MiniLM	llama3.1-8b	feedback
3 ms	22 ms	48 ms	1.2 s	loop

Runtime: Tauri 2 shell · Rust sidecar · React 19 frontend · Ollama (llama3.1-8b) · SQLCipher SQLite

END-TO-END P95	P50 HYBRID SEARCH	MEMORY FOOTPRINT	DATA EXFIL
1.8s	22ms	~5GB	0 B
full draft, hybrid search + 8B token gen	TF-IDF + MiniLM-L6 rerank	llama3.1-8b q4 + app + indexes	everything stays on the machine

CHAPTER 04 · DEMO

The workspace — composer, answer, triage.



01

Composer

Paste a ticket, pick the intent chip, set response length — ⌘↵ generates a draft.

02

Hero answer

16px / 1.65 prose at 70ch. Inline [n] pills click through to the cited source.

03

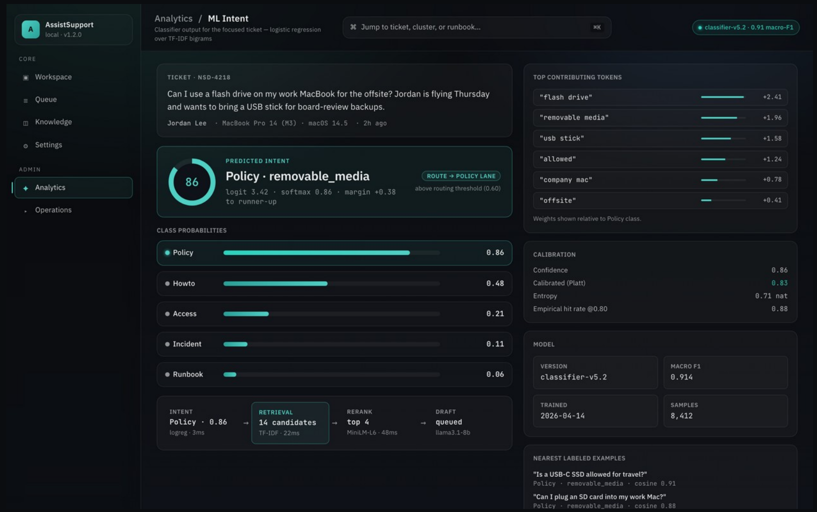
Triage rail

Workflow · signals · alternatives · feedback · context — all in one column.

CHAPTER 05 · ML INTENT

Why logreg + TF-IDF beat embeddings here.

- Logistic regression over TF-IDF bigrams — 3 ms on-device, 0.914 macro-F1 across policy / howto / access / incident / runbook.
- Calibrated with Platt scaling so the softmax score actually means what it claims — at ≥0.80 the hit rate is 0.88 empirically.
- Feature weights are inspectable: every routing decision is a ranked list of tokens, not a dense vector. Easy to debug, retrain.
- Dense embeddings would have matched F1 at ~50× the latency and ~500× the model size. Wrong tool for the budget.



Live classifier trace for NSD-4218

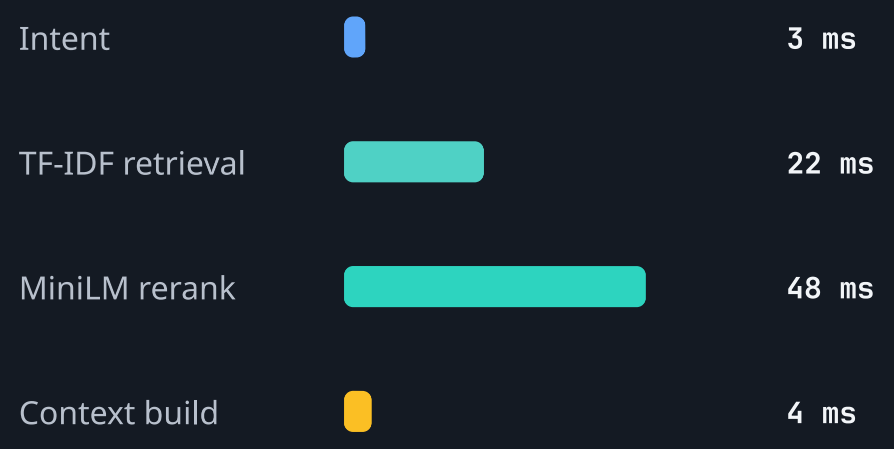
MACRO-F1	LATENCY	MODEL SIZE
0.914	3 ms	4 MB
40-case eval suite #4812	per ticket, on-device	vs 450MB+ for a small BERT

CHAPTER 06 · HYBRID SEARCH

Sub-25 ms retrieval over the local KB.

- Stage 1 — TF-IDF returns ~14 candidates in 22 ms. Cheap, deterministic, no GPU.
- Stage 2 — ms-marco-MiniLM-L-6-v2 cross-encoder reranks the candidates in 48 ms on CPU.
- Top-4 survive into the draft as the LLM's context. Each one carries a citable title + heading path.
- The reranker is the quality lever — TF-IDF alone would cite topically relevant but semantically wrong articles.

LATENCY BUDGET · p50



End to retrieval: 77 ms · then LLM draft streams in 1.2 s

P50 HYBRID SEARCH	P95 HYBRID SEARCH	DEMO KB
22ms	46ms	27
TF-IDF candidate retrieval	measured on M3 MBP	sanitized markdown sources

The model is allowed to say 'I don't know.'

ANSWER

Confidence ≥ 0.80 and all claims grounded. Draft ships with inline [n] citations.

CLARIFY

0.60–0.79 or partial grounding. The draft asks one targeted clarifying question back.

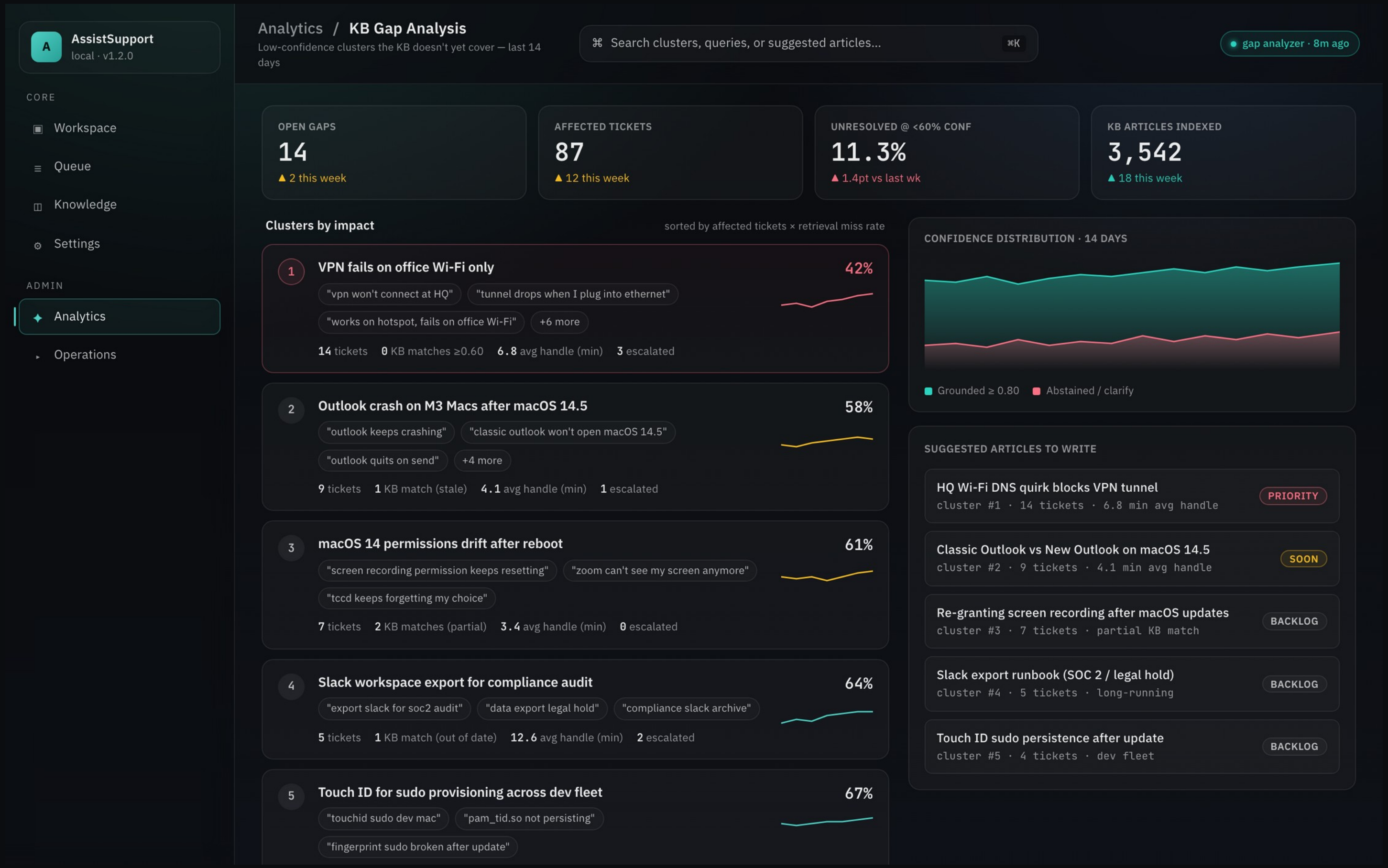
ABSTAIN

Below threshold or unsupported. Flag the ticket as a KB gap candidate and surface to the operator.

- Inline citations are generated into the prompt, not post-hoc — so the model can't cite a doc it didn't see.
- Grounded-claims check runs a per-sentence match against retrieved chunks; unsupported sentences get flagged.
- Operators thumbs-up / thumbs-down every draft. Thumbs-down feeds straight into the KB gap analyzer (next slide).

CHAPTER 08 · FEEDBACK LOOP

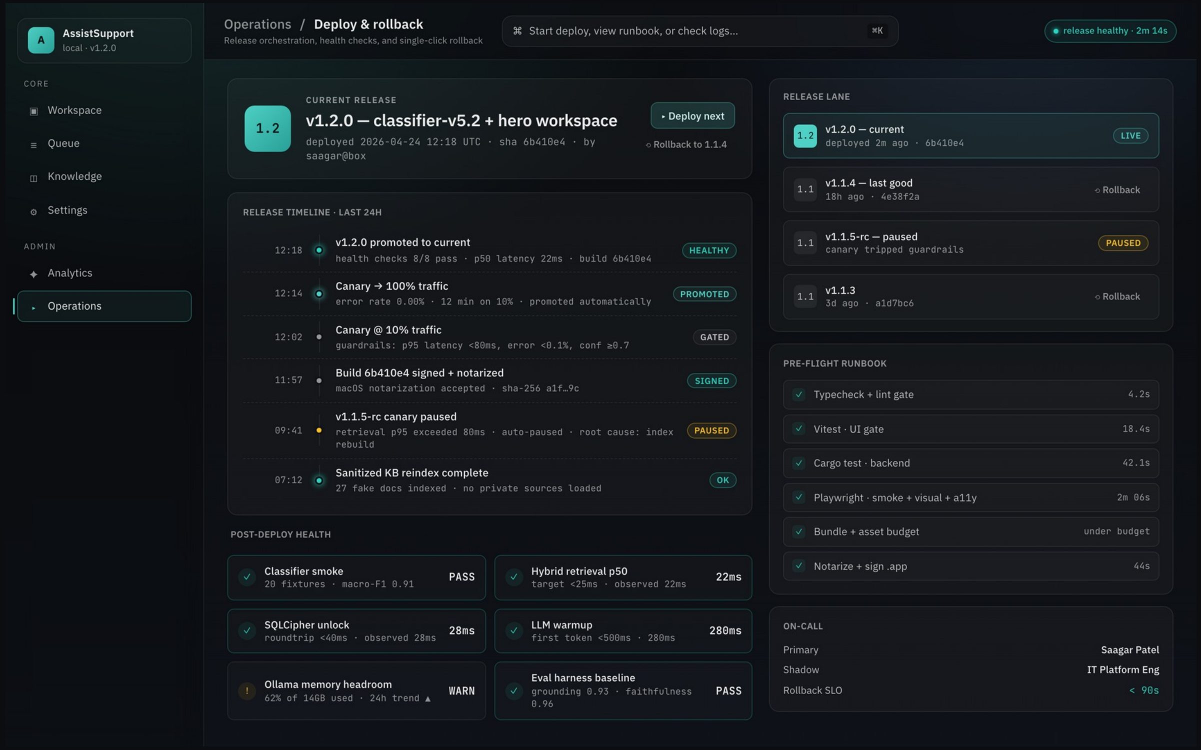
Low-confidence queries become the KB backlog.



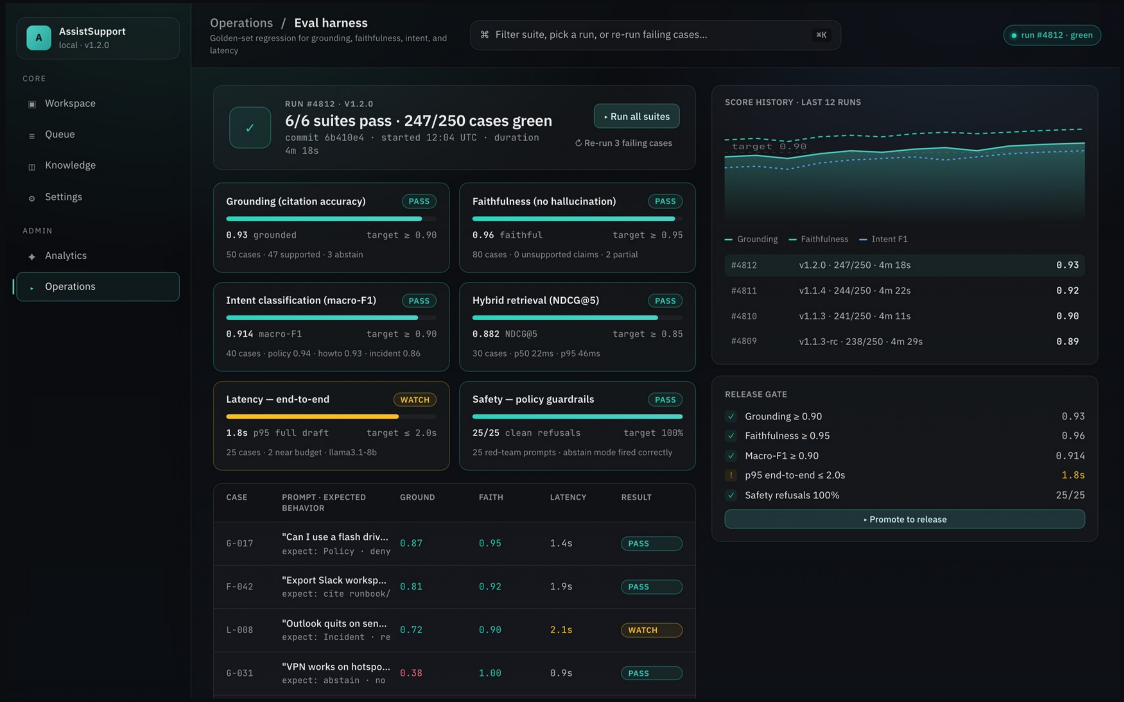
- Every abstained or low-confidence query lands in a cluster.
- Clusters ranked by impact = affected tickets × retrieval miss rate.
- Top clusters become a prioritized list of KB articles to write.
- Writers fill the gap → next week's confidence distribution shifts right.
- The loop is measurable: 14-day view shows grounded-vs-abstained trend.

CHAPTER 09 · OPS

Yes, a desktop app needs a deploy story.



Deploy / rollback surface



Eval harness · run #4812

Canary on 10% → guardrails on p95 latency, error rate, and grounding score → auto-promote. 90-second rollback SLO.

Five things I didn't expect.

01 **Local-first is a UX decision, not just a security one.**

Operators trust a tool more when they can literally turn their Wi-Fi off and it still works. The privacy story lands emotionally.

02 **Prompt-cache hits are the real latency win.**

The intent + retrieval output is cached per-ticket — second generations are 3× faster. Worth more than model quantization.

03 **Logreg is not a downgrade — it's a feature.**

Inspectable weights mean every routing decision is defensible. 'Why did you send this to the policy lane' has a concrete answer.

04 **Tauri + Rust is the right desktop stack in 2026.**

Bundle size, Apple notarization, and Rust FFI for the ML sidecar made iteration 2-3× faster than the Electron alternative.

05 **The feedback loop only works if rating is one click.**

Anything longer than thumbs-up / thumbs-down gets skipped. All the KB gap data comes from that single-click surface.

THANKS FOR WATCHING

Questions?

Open source · MIT licensed · runs on any M-series MacBook with Ollama installed.

REPO

github.com/saagpatel/AssistSupport

current repo · v1.2.0 · MIT

DECK + ONE-PAGER

portfolio drop

PDF + slide deck + screenshot set

CONNECT

in/saagarpatel

DMs open — IT platform + local AI

Built with Tauri 2 · React 19 · TypeScript · Rust · SQLCipher · Ollama · TF-IDF + MiniLM-L-6-v2 · logreg intent classifier